ORIGINAL RESEARCH ARTICLE

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Comparative Study of the Effects of Two Once-a-Month Injectable Steroidal Contraceptives (Mesigyna® and Cyclofem®) on Glucose Metabolism and Liver Function

United Nations Development Programme/United Nations Population Fund/World Health Organization/World Bank, Special Programme of Research, Development and Research Training in Human Reproduction, Task Force on Long-acting Systemic Agents for Fertility Regulation*

A 1-year, longitudinal, multicenter study was undertaken to monitor the effects of two once-a-month, injectable contraceptive preparations, Mesigyna® (norethisterone enanthate, 50 mg, and estradiol valerate, 5 mg) and Cyclofem® (medroxyprogesterone acetate, 25 mg, and estradiol cypionate, 5 mg) on carbohydrate metabolism and liver function. A total of 357 volunteers from four centers (Hangzhou, Havana, Jakarta, and Shanghai) were monitored during one pretreatment cycle, at the end of the ninth injection interval and during one posttreatment cycle, approximately 3 months after the 9-month period of treatment was completed. Both preparations induced minor changes in glucose tolerance, with a slight elevation of glucose response and minimal changes in insulin response to a glucose challenge. These changes were greater with Mesigyna but were of no clinical relevance in either group. With regard to liver function, both preparations induced a rise in bilirubin, a decrease in alkaline phosphatase, and no change in enzyme serum levels. However, these changes were of small magnitude and levels remained well within the normal range in all subjects. CONTRACEPTION 1998;57: 71–81 © *1998 Elsevier Science Inc. All rights reserved.*

KEY WORDS: Cyclofem®, Mesigyna®, norethisterone enanthate, medroxyprogesterone acetate, estradiol valerate, estradiol cypionate, carbohydrate metabolism, liver function

Introduction

Two new combined, once-a-month injectables were developed by the United Nations Development Programme/United Nations Population Fund/World Health Organization/World Bank Special Programme of Research, Development and Research Training in Human Reproduction, which are now available to family planning programs: Cyclofem® (medroxyprogesterone acetate, 25 mg, and estradiol cypionate, 5 mg) and Mesigyna® (norethisterone enanthate, 50 mg, and estradiol valerate, 5 mg).

Compared with progestogen-only injectable preparations, both were found to be equally effective methods[1] and to be more acceptable by inducing fewer menstrual disturbances.[2]

Ever since it was recognized that steroid contraceptive preparations had effects on carbohydrate and lipid metabolism and that this had an impact on the risk of users for cardiovascular disease, research has aimed at developing methods that have minimal metabolic effects. Thus, as part of the assessment of the safety of these two once-a-month injectables, a multicenter longitudinal study was conducted to monitor their metabolic effects. The data on their effects on lipid metabolism are published in a separate report.[3] This article reports the findings of their effects on glucose metabolism and liver function, monitored during the same clinical trial.

Materials and Methods

Subjects

Subjects were recruited in four centers: Hangzhou, Havana, Jakarta, and Shanghai. They were healthy,

*For members, see Appendix.

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informed, female volunteers, aged 18–35 years (inclusive). They had to have had at least two regular cycles since stopping their last contraceptive method and to have stopped lactating for at least 6 months before admission to the study. Postpartum and postabortion subjects had to be at least 6 mo postpartum or postabortion and to have had one normal cycle since delivery or abortion.

Women were not enrolled if they had diabetes; a Papanicolaou smear grade 3, 4, or 5; a history of thromboembolism, vaginal bleeding of unknown etiology, hypertension, hyperlipidemia, recent severe liver disease, a known or suspected malignancy, an abnormal nipple discharge, or low hemoglobin by local standards; if they were nulliparous or pregnant; if they used any of the following drugs: barbiturates, anticonvulsants, rifampicin, systemic corticosteroids, drugs affecting the cardiovascular or hepatic systems, or any drug used on a long term basis; if they had taken oral contraceptives within 3 months or had had injectable contraceptives administered within 6 months of admission. Obese women, defined as having a body mass index >25 kg/m^2, and smokers were also excluded.

Study Design

Details of the study design were published previously.[3] Of relevance to the data presented here are the following aspects of the study. After screening, volunteers were asked to return to the clinic during an untreated menstrual cycle (pretreatment) for venipuncture. They were then randomly allocated to one of the two treatment groups. The first injection was given in the first 5 days of the next menstrual cycle, followed by eight more injections at monthly intervals. At the end of the period of contraceptive protection, ie, 4 weeks after the ninth and final injection, venipuncture was repeated. Volunteers were then asked to use barrier methods of contraception for 3 months and venipuncture was again performed during the third month (posttreatment).

Laboratory Tests

On each occasion that venipuncture was scheduled, a 2-h oral glucose tolerance test (OGTT) with a 75-g oral glucose challenge was performed. Blood samples were obtained at fasting and every 30 min until 2 h after glucose ingestion, and subsequently assayed for serum glucose and insulin levels. In addition, the fasting blood sample was assayed for the following liver function tests: alanine transaminase (SGPT), aspartate transaminase (SGOT), γ-glutamyl transpeptidase (γGT), alkaline phosphatase, and bilirubin.

Laboratories assayed the blood samples using their routine clinical chemistry methods. All used enzymatic methods[4,5] for the glucose assay and the same methods for the insulin,[6] γGT,[7] alkaline phosphatase,[8] and bilirubin[9] assays. For the transaminases, all laboratories used the method recommended by the International Federation of Clinical Chemistry (IFCC),[10,11] except in Shanghai where the Reitman-Frankel method[12] was used. All laboratories followed strict internal quality control procedures.

Statistical Analysis

The number of subjects with abnormally high glucose values at one or more of the five measurements was determined by treatment group and center, pretreatment, at the end of the treatment period, and posttreatment. Serum glucose was considered abnormally high if it was above 6.4, 11.1, 11.1, 11.1, and 7.8 mmol/L at fasting, and at 30, 60, 90, and 120 min after the glucose challenge, respectively. The percentage of subjects with at least one abnormally high value in each follow-up was compared with that at admission using the McNemar test. The 95% confidence intervals (CI) for the percentages were determined using the binomial distribution.

For all centers, at each visit, repeated measures analysis was used with glucose values to obtain means, standard errors, and 95% confidence intervals by treatment group at fasting, and at 30, 60, 90, and 120 min. The means were plotted to show the combined, center-adjusted OGTT glucose curves for each group at each visit. Repeated measures analysis of variance was also used to calculate the within-center means for both glucose and insulin, and the means were plotted to show the mean OGTT glucose and insulin curves for each center.

The differences in glucose and insulin values between each time after the glucose challenge and fasting were calculated for each subject. Repeated measures analysis of variance was used to compare these changes between the end of treatment and pretreatment and between posttreatment and pretreatment for each treatment, and also between treatments at each visit. In the case of insulin, the comparisons were made only within centers.

An approximation of the area under the insulin curve was calculated for each subject as $0.5 \times [0.5 \times$ (Value at fasting) + Value at 1/2 h + Value at 1 h + Value at 1 1/2 h + $0.5 \times$ Value at 2 h]. The ratios of the areas under the insulin curve were then computed for each subject between the post- and pretreatment visits, between the end of treatment and the pretreatment visits, and between the posttreatment and the end of treatment visits. After applying the logarithmic transformation, adjusted means were calculated

with their 95% confidence intervals and the significance of the changes was assessed using the *t* test for within-center comparisons.

For liver function tests, the number of subjects with abnormally high values at each visit was determined by treatment group and center, at the pretreatment, end of treatment, and posttreatment visits. The percentage of abnormally high values at each follow-up was compared with that at admission using the McNemar test. The ratios of the liver function values were computed for each subject between the end of treatment and the pretreatment visits and between the posttreatment and the pretreatment visits. The following analyses were then performed for each center, using logarithmic transformation: paired *t* tests to assess the significance of each change and *t* tests for independent samples to compare the change between the two treatment groups. The mean ratios in the logarithmic scale and their 95% confidence intervals were also computed and the retransformed values were multiplied by the admission values to draw the graphs in the original scale.

Results

Numbers

Data from 357 subjects (184 Mesigyna users and 173 Cyclofem users) were included in the analysis. Of those, 154 Mesigyna users and 144 Cyclofem users completed the treatment period of follow-up and 142 in each group completed the full period of follow-up. Characteristics of the study volunteers at admission, as well as reasons for exclusions and reasons for discontinuation were described in a previous publication.[3]

Glucose Metabolism

All subjects had a normal OGTT at admission. At the end of the treatment period and 3 months after treatment, there were a few subjects with an abnormal OGTT curve (see later here), but in neither treatment group were the increases in the percentages of subjects with abnormal values relevant or significant.

In the Cyclofem group, only one of 144 subjects (0.7%, 95% confidence interval: 0.0–3.8%) had an abnormal OGTT curve at the end of the treatment period, with a slightly elevated serum fasting glucose level (6.5 mmol/L) and normal glucose levels during the remainder of the test. Three months posttreatment, another subject, one of 142 Cyclofem users (0.7%, 95% confidence interval: 0.0–3.9%), had a slight abnormality in her response to the glucose challenge, with an elevated serum glucose level of 9.8 mmol/L at 2 h.

In the Mesigyna group, only one of 154 subjects (0.7%, 95% confidence interval: 0.0–3.6%) had an abnormal OGTT curve at the end of the treatment period, with an elevated serum fasting glucose level (6.9 mmol/L) and normal glucose levels during the remainder of the test. Three months posttreatment, four of 142 Mesigyna users (2.8%, 95% confidence interval: 0.8–7.1%) had an abnormal OGTT curve: two had an elevated fasting serum glucose (6.6 and 6.7 mmol/L); one also had a serum glucose of 8.3 mmol/L at 2 h, above the normal range for normal glucose tolerance. Two subjects had a single elevated serum glucose level; for one, this was at 1 h (12.5 mmol/L) and for the other, at 2 h (8.8 mmol/L).

Table 1 shows the center-adjusted glucose means and their 95% confidence intervals by treatment group at fasting (0 min) and at 30, 60, 90, and 120 min after the 75-g oral glucose challenge, before treatment, at the end of the 9-month treatment period, and 3 months posttreatment, for all centers combined. Figure 1 presents these means in graphic form, showing the significance of the differences in the changes from fasting levels between the end of treatment and pretreatment and between posttreatment and pretreatment. The response of glucose values to the glucose challenge was increased during treatment, mainly among Mesigyna users. However, the maximal magnitude of the increase, which occurred at the end of the treatment period among Mesigyna users at 90 and 120 min, was 0.53 mmol/L (95% confidence intervals: 0.37–0.69 and 0.37–0.68 mmol/L, respectively), of no clinical relevance, the upper 95% confidence limit for the mean being always well below the threshold for abnormally high values (see Table 1). Moreover, there was some recovery after treatment, with the serum glucose levels shifting back in the direction of the pretreatment values. The shape of the curves was similar across centers (interactions by centers nonsignificant at 5%).

The increase in glucose response to the glucose challenge with Mesigyna during treatment tended to be larger than with Cyclofem. Although at 30 and 60 min there was a significant increase among Mesigyna users with respect to pretreatment levels, there was practically no change among Cyclofem users ($p = 0.0198$ and $p = 0.0100$ for the differences between the changes in the two groups at 30 and 60 min, respectively). At 90 and 120 min there was an increase in both groups, but the increase for Mesigyna users was higher than for Cyclofem users ($p = 0.0150$ and $p = 0.0239$, respectively). After treatment, means among Mesigyna users were still above the pretreatment levels at 120 min (0.23 mmol/L above, 95% confi-

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Table 1. Glucose values at fasting (0 min), and at 30, 60, 90, and 120 min after glucose intake pretreatment, at the end of the treatment period, and post-treatment (by treatment group, all centers, means and 95% confidence intervals [CI])

		Pretreatment						End of Treatment						Posttreatment						
		Cyclofem			Mesigyna			Cyclofem			Mesigyna			Cyclofem			Mesigyna			
	Minutes	Lower 95% CI	Mean	Upper 95% CI	Lower 95% CI	Mean	Upper 95% CI	Lower 95% CI	Mean	Upper 95% CI	Lower 95% CI	Mean	Upper 95% CI	Lower 95% CI	Mean	Upper 95% CI	Lower 95% CI	Mean	Upper 95% CI	Upper Threshold for Normal
Glucose (mmol/L)	0	4.31	4.39	4.48	4.28	4.37	4.45	4.31	4.40	4.49	4.38	4.47	4.56	4.30	4.39	4.49	4.26	4.35	4.45	6.4
	30	5.99	6.15	6.30	5.90	6.05	6.20	5.98	6.14	6.29	6.18	6.33	6.48	5.97	6.16	6.34	5.86	6.04	6.22	11.1
	60	5.77	5.94	6.10	5.58	5.74	5.90	5.76	5.94	6.12	5.96	6.13	6.30	5.61	5.82	6.03	5.73	5.94	6.15	11.1
	90	5.09	5.22	5.35	5.02	5.15	5.27	5.27	5.42	5.58	5.49	5.64	5.79	5.19	5.37	5.55	5.12	5.30	5.48	11.1
	120	4.71	4.83	4.94	4.59	4.71	4.82	4.93	5.08	5.22	5.08	5.22	5.36	4.80	4.95	5.11	4.75	4.91	5.06	7.8

dence interval 0.06, 0.41 mmol/L), whereas means among Cyclofem users were closer to admission levels.

Figure 2 shows the glucose means by treatment group at the different times after the glucose challenge, for each visit, by center. Subjects from Havana had glucose mean values lower than those of the other centers: the pretreatment values at 0 min were 3.3 mmol/L (standard deviation [SD] 0.8) and 3.2 mmol/L (SD 0.9) for Cyclofem and Mesigyna, respectively, whereas in the other centers combined, they were 4.8 mmol/L in both groups (SD 0.4). The peak response was observed at 30 min except in Jakarta, where it occurred at 60 min.

Figure 3 shows the mean insulin values by treatment group at different times after the glucose challenge, for each visit, by center, except Jakarta, for which there were no insulin data available. The values were much lower in Hangzhou than in the other centers. Although in Hangzhou and Shanghai the OGTT curves looked similar and the same response was observed between visits, in Havana the mean values after the glucose challenge were much higher before treatment than at the end of the treatment period and posttreatment.

Table 2 shows the means and their 95% confidence intervals of the percentage changes in the areas under the insulin curve (AUIC) between the end of the treatment period and pretreatment, posttreatment and pretreatment, and posttreatment and the end of the treatment period, by treatment group and center. The p values for testing the hypothesis of no change, comparing changes between treatment groups, are also shown by center. With Cyclofem, there was no effect observed in either Hangzhou or Shanghai, and a significant decrease in AUIC was observed during and after treatment in Havana. With Mesigyna, in each center, there was a different pattern: in Hangzhou, there was an increase of 24% in posttreatment AUIC with respect to pretreatment, whereas in Havana there was a decrease to 67% at the end of treatment and to 62% posttreatment, and in Shanghai there was a smaller decrease 3 months posttreatment.

Liver Function Tests

The occurrence of abnormally high values of liver functions at admission, taking as cut-off points center-specific values, were 28 (7.8%), 3 (0.8%), 4 (1.1%), 1 (0.3%), and 6 (1.7%) for SGOT, SGPT, γ-glutamyl transpeptidase, alkaline phosphatase, and bilirubin, respectively. The 28 high values for SGOT all came from Havana. There were no elevated values in Jakarta. Among the six high values for bilirubin, four came from Hangzhou.

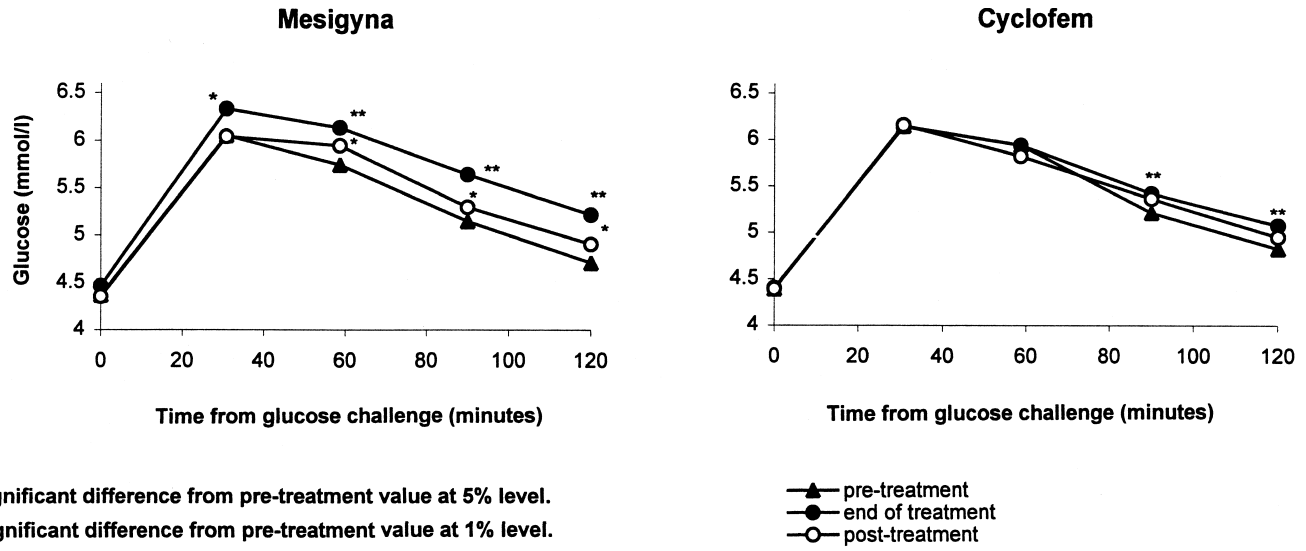

Figure 1. Oral glucose tolerance test (OGTT) pretreatment, at the end of the treatment period, and posttreatment for Cyclofem and Mesigyna users, in all centers.

With treatment, there was only one significant increase in the percentage of abnormally high values, occurring in bilirubin among Cyclofem users, from 1 (0.6%) at admission to 9 (6.6%) at the end of the treatment period. However, this was the only significant result at 5% level in 20 tests performed, suggesting that this might have occurred by chance.

Figure 4 presents the mean values of the five liver functions before treatment, at the end of the 9-month treatment period and 3 months posttreatment, by center and treatment group. Fluctuations in liver functions were small and differed across centers. In the group of Cyclofem users, a decrease in SGOT was observed in Hangzhou only, at the end of the treatment period with partial recovery posttreatment. Changes in SGPT and γGT were observed in two centers in opposite directions, with rises in both levels in Jakarta and falls in both in Shanghai, in each case observed at the end of the treatment period and maintained posttreatment. The two most consistent findings were a drop in alkaline phosphatase and a rise in bilirubin, observed in all centers but Jakarta, at the end of the treatment period and continuing posttreatment.

In the group of Mesigyna users, a small rise in SGOT was observed in Jakarta at the end of the treatment period, which had reverted to baseline 3 months posttreatment, whereas an increase in this enzyme serum level was noted in Hangzhou posttreatment. A rise in SGPT was noted in Jakarta at the end of treatment with partial recovery posttreatment, whereas a small drop in this enzyme serum level was observed in Shanghai at the end of the treatment and was maintained posttreatment. Rises in γGT were observed in Hangzhou and Jakarta at the end of the treatment period, with partial recovery posttreatment. As in the case of Cyclofem, the most consistent finding was a decrease in alkaline phosphatase in all centers but Jakarta observed at the end of treatment and further pronounced posttreatment, as well as a rise in bilirubin observed in all centers at the end of the treatment period with partial recovery posttreatment.

When the comparison of both treatments was performed, significant differences were only observed in Hangzhou, where relatively greater changes in γGT at the end of treatment and SGOT posttreatment were observed in the Mesigyna group, compared with the Cyclofem group. In fact, the greatest contrast was seen at the end of the treatment period, when Mesigyna had a 61% rise and Cyclofem a 11% drop in γGT, compared with pretreatment levels.

Discussion

Ever since the late 1960s, when the effects of combined oral contraceptives on carbohydrate metabolism were first recognized, numerous studies have been conducted to assess the effects of the various synthetic steroids contained in different preparations. Early on, the effect was thought to be due to the estrogenic component, but then it was shown that estrogens have a biphasic effect. High doses of estrogen will cause deterioration of carbohydrate metabolism with hyperinsulinemia, together with a decrease and a delay in the early insulin response to a glucose challenge.[13] Small amounts of estrogens as found in current oral contraceptive preparations improve car-

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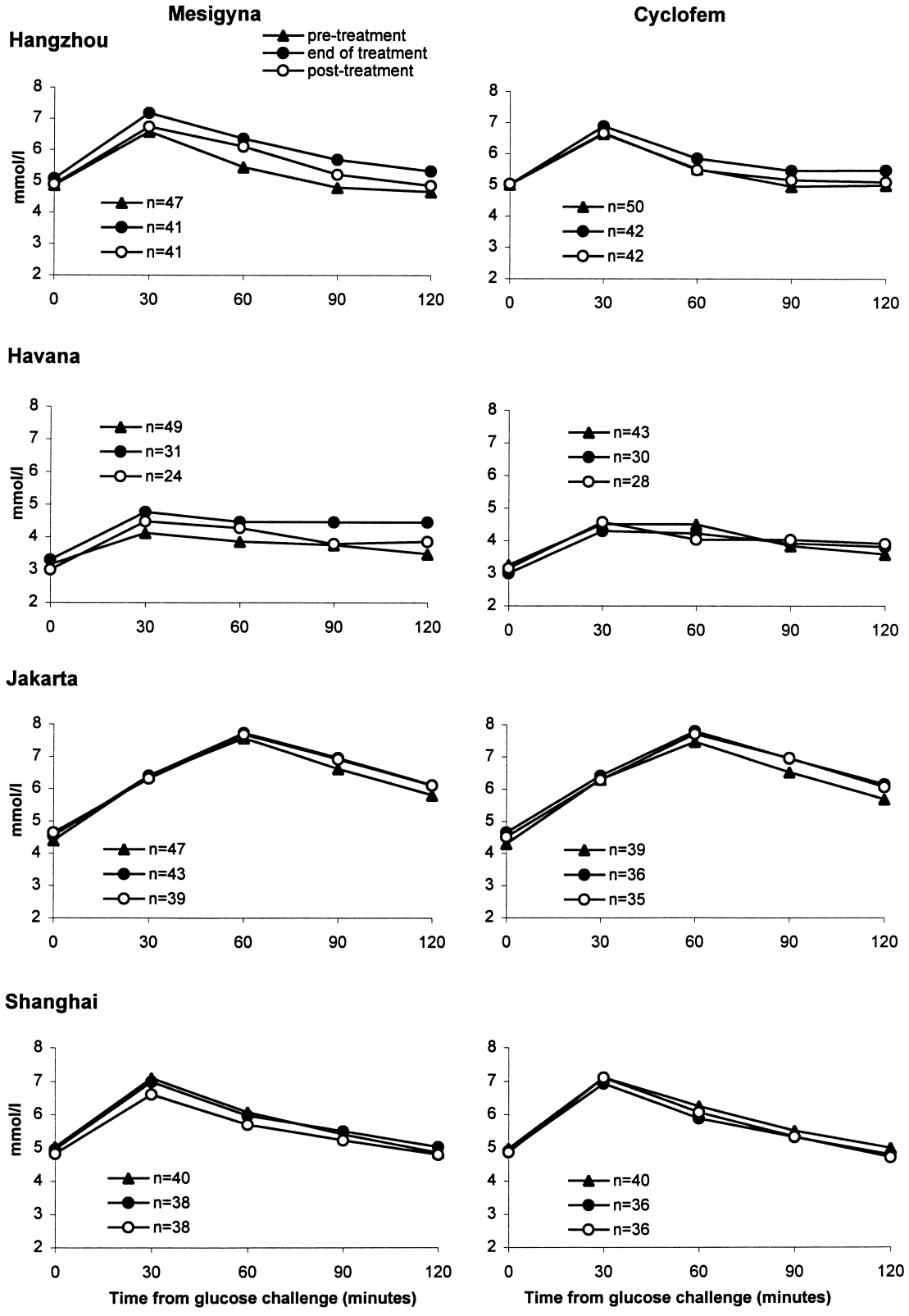

Figure 2. Oral glucose tolerance test (OGTT) pretreatment, at the end of the treatment period, and posttreatment for Cyclofem and Mesigyna users, by center: serum glucose (mmol/L).

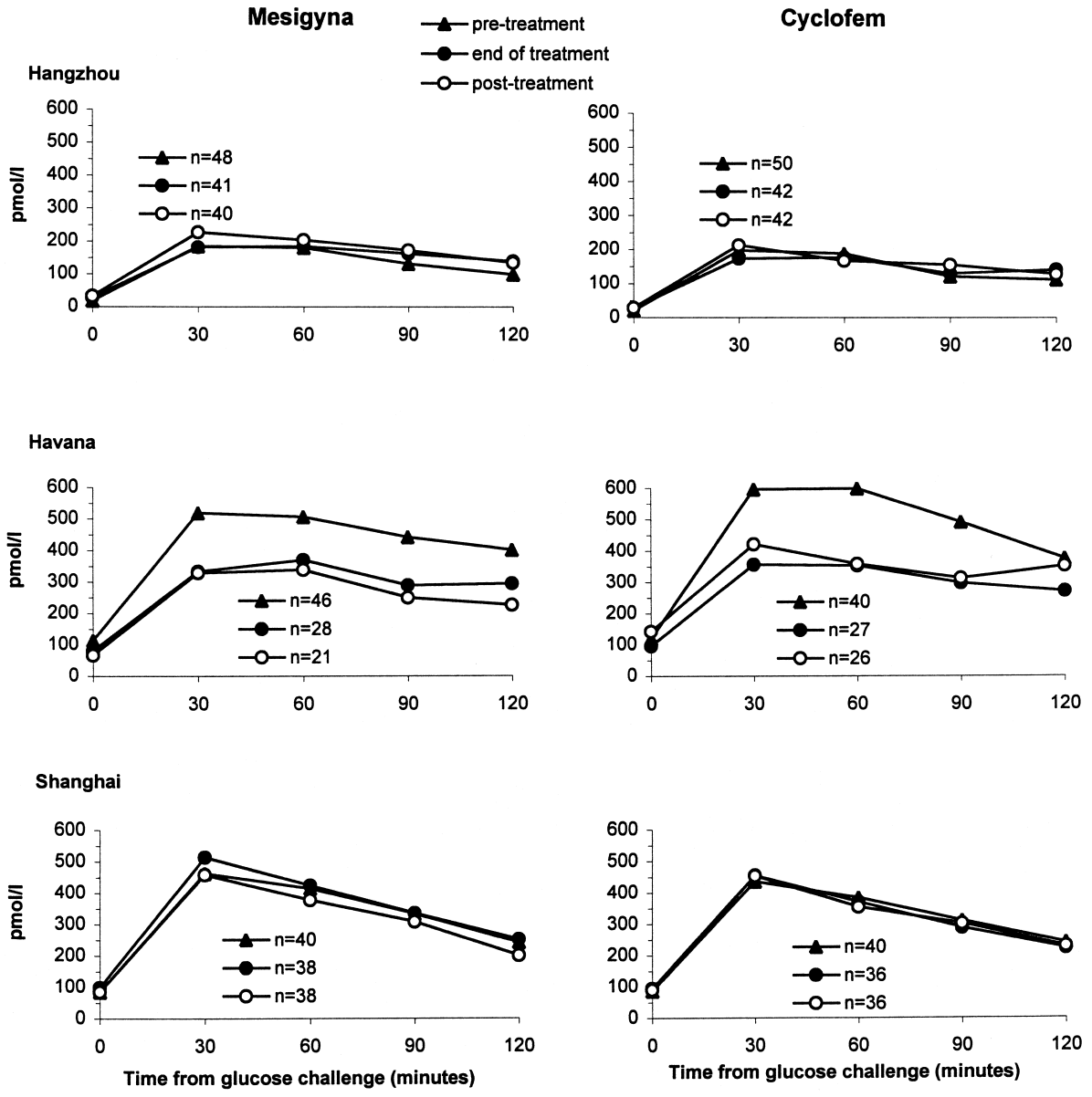

Figure 3. Oral glucose tolerance test (OGTT) pretreatment, at the end of the treatment period, and posttreatment for Cyclofem and Mesigyna users, by center: insulin response (pmol/L).

bohydrate metabolism by alteration of insulin receptor binding and inhibition of insulin degradation.[14]

The progestogenic component of low dose oral contraceptives was found to have the most effect on carbohydrate metabolism. Progestins induce an impairment in glucose tolerance with hyperinsulinemia. This reflects both an increased pancreatic β-cell response to glucose[15] as well as an increased insulin resistance due to a decreased affinity of insulin for its receptor or to a decreased amount of insulin receptors on cell membranes.[16] These effects are dependent on the type of progestins, gonanes having more effect than estranes or pregnanes, and are dose-related. In fact, they are similar to those induced by progesterone, which have been well documented.[17]

More recently, it was demonstrated that hyperglycemia and hyperinsulinemia were independent risk factors in cardiovascular disease[18,19] and that insulin resistance was associated with an increased risk of coronary artery disease, particularly in women.[20,21] In fact, there is evidence that the arterial wall is an insulin-sensitive tissue and that chronic exposure to high levels of insulin results in the development of lesions similar to those of early atherosclerosis.[22] In this context, it is particularly important that new contraceptive methods such as

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Table 2. Percent change in area under the insulin curve between visits (means, 95% confidence intervals [CI], and p values)

		Treatment Group								
		Cyclofem				Mesigyna				Difference Between Groups, p Value
Center	Difference between	Lower 95% CI	Mean	Upper 95% CI	p Value	Lower 95% CI	Mean	Upper 95% CI	p Value	
Hangzhou	End of treatment and pretreatment	80.6	95.2	112.4	0.57	87.2	105.3	127.3	0.59	0.43
	Posttreatment and pretreatment	84.1	100.6	120.3	0.95	105.1	123.8	145.8	0.01	0.10
	Posttreatment and end of treatment	90.4	107.5	127.9	0.42	98.8	117.9	140.8	0.08	0.47
Havana	End of treatment and pretreatment	51.3	67.4	88.5	0.01	52.7	67.3	86.0	0.01	0.99
	Posttreatment and pretreatment	58.6	72.2	89.1	0.01	48.1	61.7	79.1	<0.01	0.41
	Posttreatment and end of treatment	82.6	107.4	139.6	0.60	73.0	91.1	113.6	0.42	0.36
Shanghai (all)	End of treatment and pretreatment	90.6	98.7	107.5	0.77	101.2	107.7	114.6	0.03	0.11
	Posttreatment and pretreatment	94.1	99.0	104.2	0.71	92.0	95.5	99.0	0.02	0.25
	Posttreatment and end of treatment	92.6	100.3	108.7	0.94	83.2	88.7	94.5	<0.01	0.02

the once-a-month injectable contraceptives should have the minimum effect on carbohydrate metabolism.

In this study, treatment with Mesigyna was found to induce a small rise in glucose response to glucose challenge, which was still maintained after treatment but was well within the normal range for glucose tolerance. With Cyclofem, the changes in glucose response were even less marked. However, the data differed markedly among centers. Pretreatment, there were center differences in fasting serum glucose and glucose and insulin responses to glucose challenge, probably reflecting the different nutritional status of these populations as well as previously documented ethnic differences in glucose metabolism.[23] At the end of the treatment period, results from Havana differed markedly from those of other centers and appear to have reflected the nutritional stress to which the local population became subjected while the trial was ongoing. In the other centers, the changes were minimal, mostly similar in both treatment groups, and suggest that both Cyclofem and Mesigyna have no clinically relevant effect on glucose metabolism.

This finding is in contrast to progestogen-only injectable contraceptives. A number of studies have been published on the effects of long term use of depot-medroxyprogesterone acetate (DMPA) on carbohydrate metabolism. These studies used different designs and methodologies and generally involved only small numbers of women, but the vast majority reported significant hyperinsulinism and some described abnormalities in glucose response in a percentage of users, during oral glucose tolerance test.[24–30] The study that is most comparable with this one is a longitudinal study of DMPA users, reported by Spellacy and coworkers.[24] They found that, of 29 women who had a normal glucose tolerance curve at baseline, 62% had a ''borderline abnormal'' or ''abnormal'' curve after 1 year of use of this injectable contraceptive. Fewer data are available on the effects of treatment with norethisterone enanthate (NET-EN) on carbohydrate metabolism, but generally the changes observed are similar to those induced by DMPA.[31,32] Thus, Cyclofem and Mesigyna appear to induce fewer changes in glucose metabolism than do DMPA or NET-EN, in consonance with the lower dose of progestin that these preparations contain.

Effects of estrogens on liver metabolism have been well documented.[33,34] They decrease bile flow and decrease bilirubin maximal secretion rate, thus allowing it to diffuse back into the blood. Certain progestins can also induce intrahepatic cholestasis, as seen with estranes and gonanes but not pregnanes. Progestins but not estrogens can cause increases in serum transaminases.

In this study, changes in transaminases observed during treatment were of small magnitude, remained well within the normal range, and lacked consistency

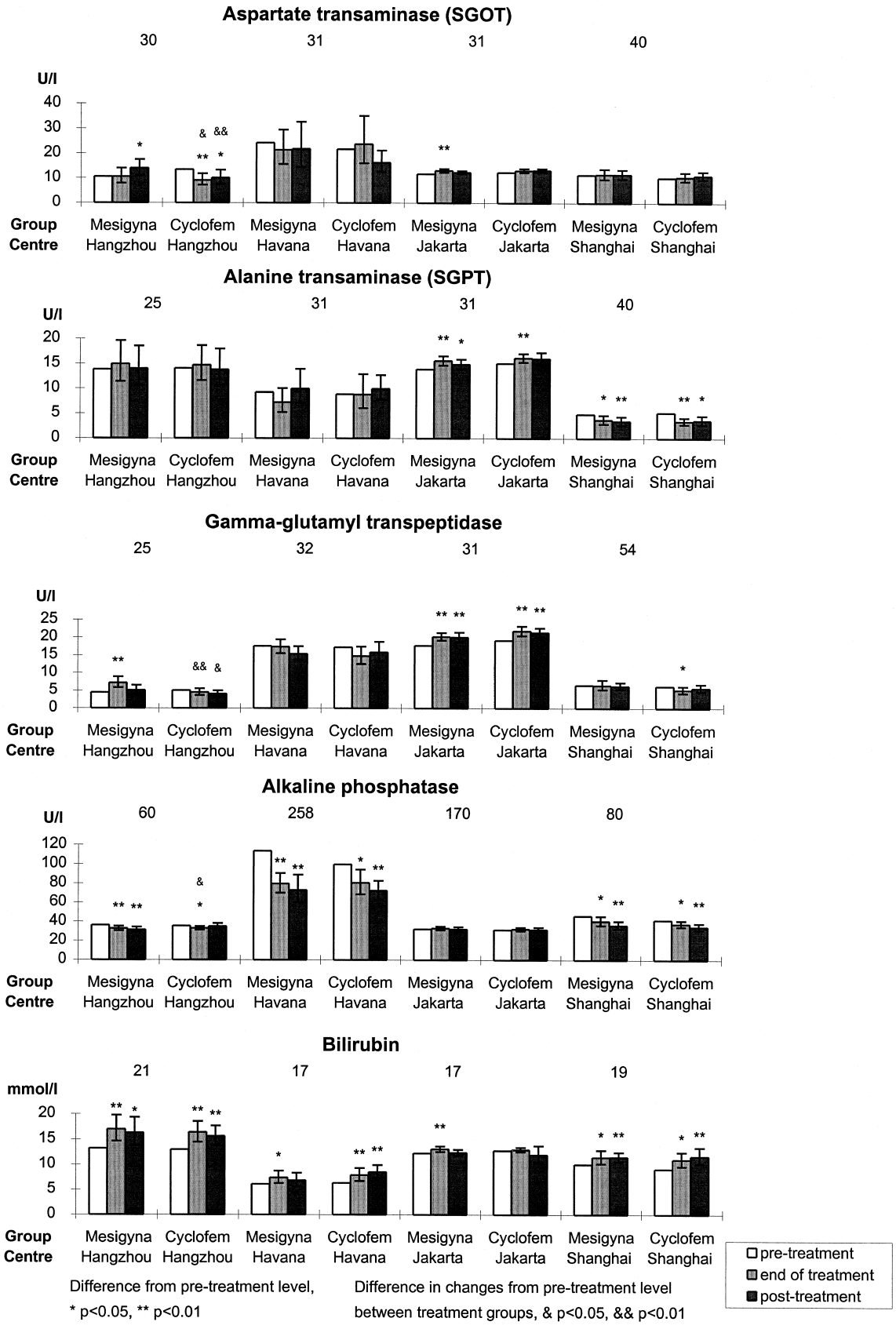

Figure 4. Mean values of liver functions and 95% confidence intervals adjusted for baseline, pretreatment, at the end of the treatment period, and posttreatment, by group and center. Center-specific upper limits for normal values are shown on top of plot area.

among centers. The more consistent findings were a decrease in alkaline phosphatase and an increase in bilirubin in both treatment groups, reflecting the estrogen component of each preparation. Differences between the two treatment groups were only significant in Hangzhou, where Mesigyna induced slightly more rises in liver enzymes than did Cyclofem, a contrast that may be the reflection of their different progestin component.

There are few data available on the effects of other injectable contraceptive preparations on liver metabolism. At the end of 1 year of use of DMPA or NET-EN, most studies report no change in serum liver enzymes or in bilirubin,[28] whereas some others find a significant impairment in the clearance of bromsulphtalein.[35]

In conclusion, Cyclofem and Mesigyna induce only minimal changes in glucose metabolism and liver function, which make them safer than progestogen-only injectable contraceptives, at least in this regard.

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Appendix

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